

Introduction

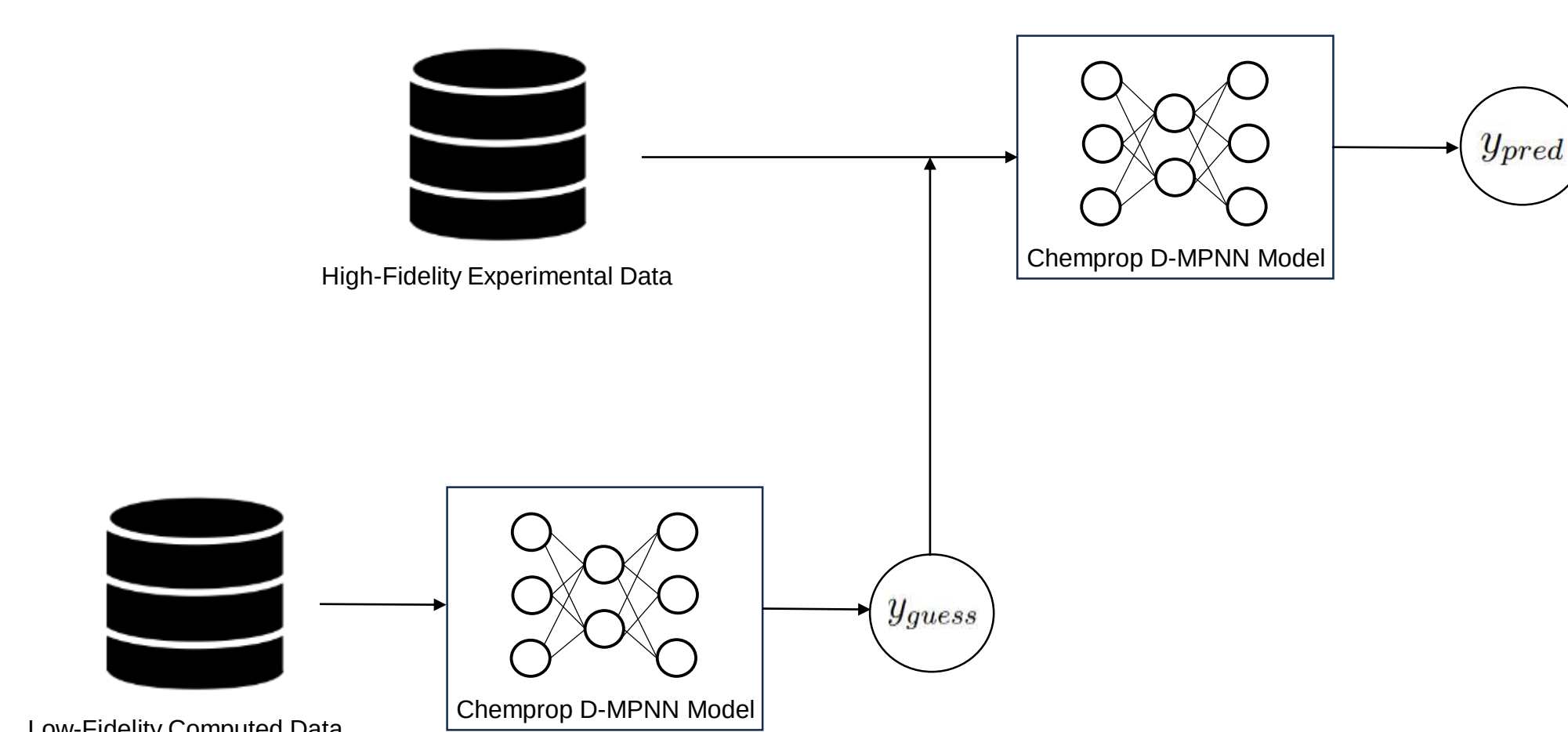
In contemporary property prediction models, the incorporation of end-to-end deep learning neural networks is commonplace. Multi-Fidelity learning provides an intermediary prediction step that in theory should help the final network learn in a data efficient manner. But simply from the low fidelity prediction it is unclear if the prediction is any good. To address this concern, we propose the integration of an evidential deep learning framework within the auxiliary neural network to forecast both aleatoric and epistemic uncertainties associated with predictions. This framework introduces a dynamic tradeoff mechanism, wherein learning predominantly relies on low fidelity predictions when uncertainties are minimal, transitioning to high fidelity labels when uncertainties become pronounced.

Models

We begin by constructing the two models for benchmarking:

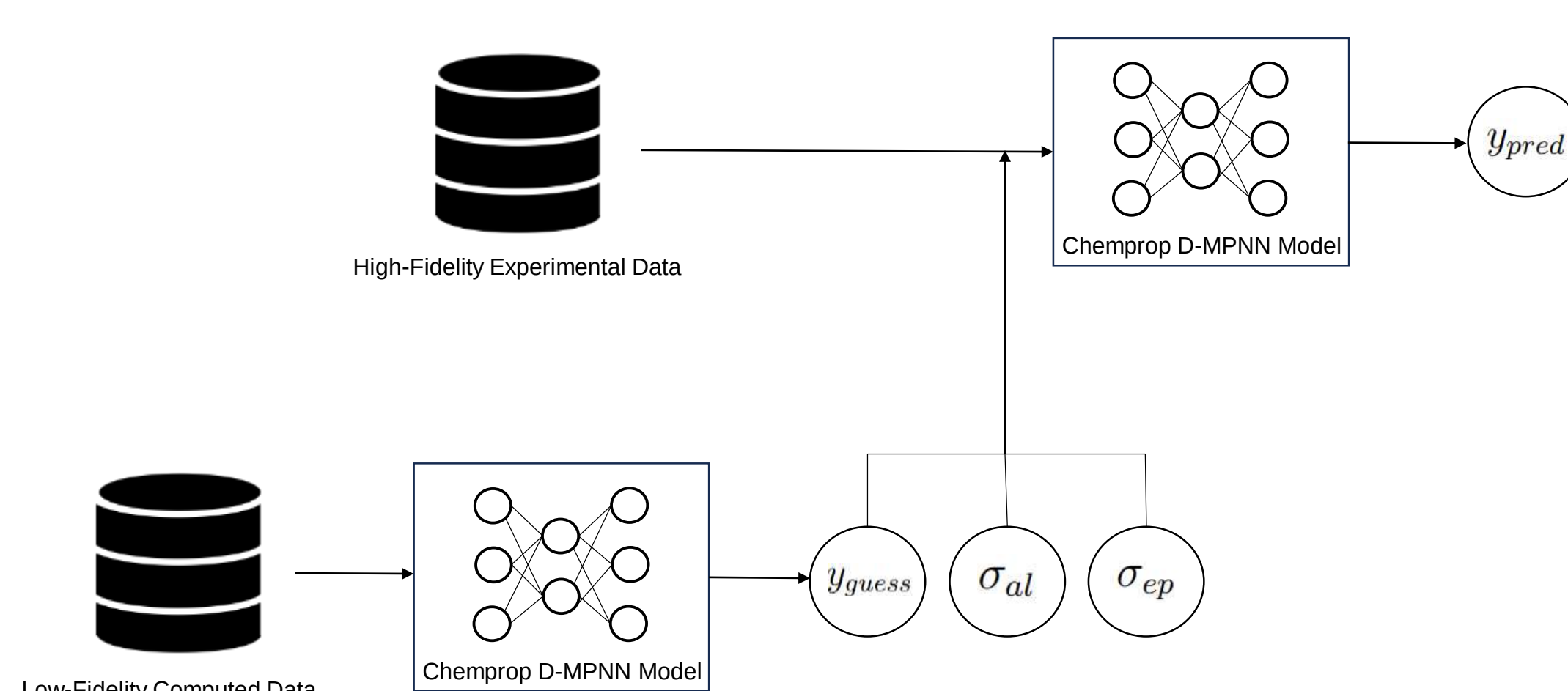
Base Multi-fidelity:

Auxiliary network learns low-fidelity targets, provides a “guess” to help with high-fidelity prediction



Evidential Multi-fidelity:

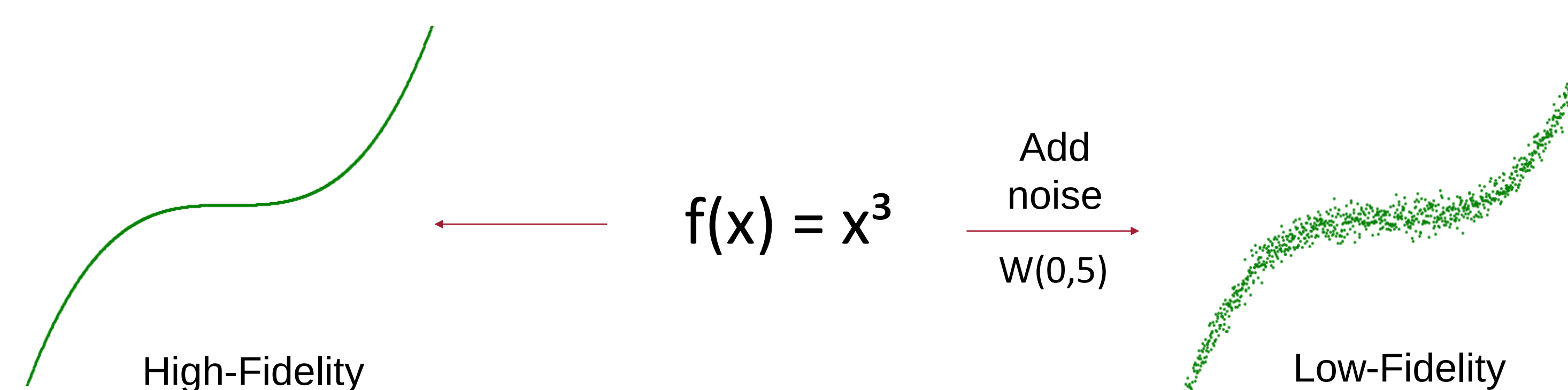
Auxiliary network learns low-fidelity targets, provides a “guess” and its confidence to help with high-fidelity prediction



Methods

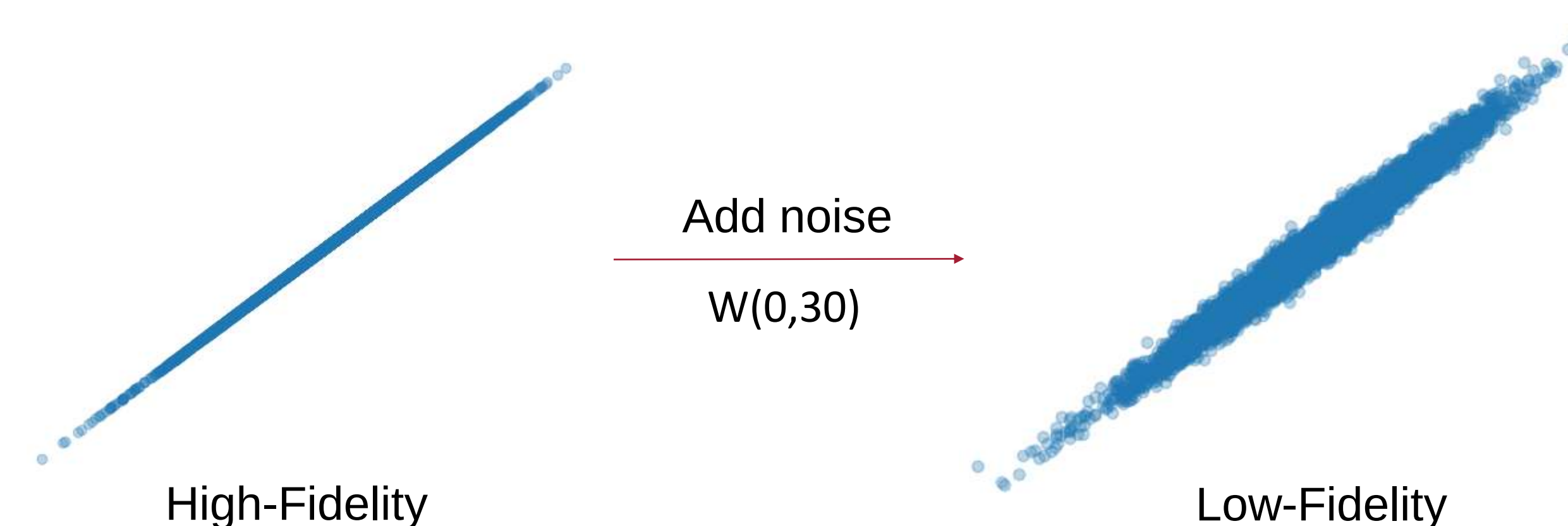
Purely Synthetic Dataset

We generate a dataset using a cubic polynomial equation with gaussian noise of standard deviation 5 and trained the model within the input range of $[-3,3]$.



Semi Synthetic Dataset

We systematically add gaussian noise with standard deviation of 30 to a large synthetic dataset of *enthalpies of formation*:



Non-synthetic Dataset

Optical properties: from experiments and time-dependent density functional theory calculations

Drug efficacy vs. potency: from single point and concentration response curves

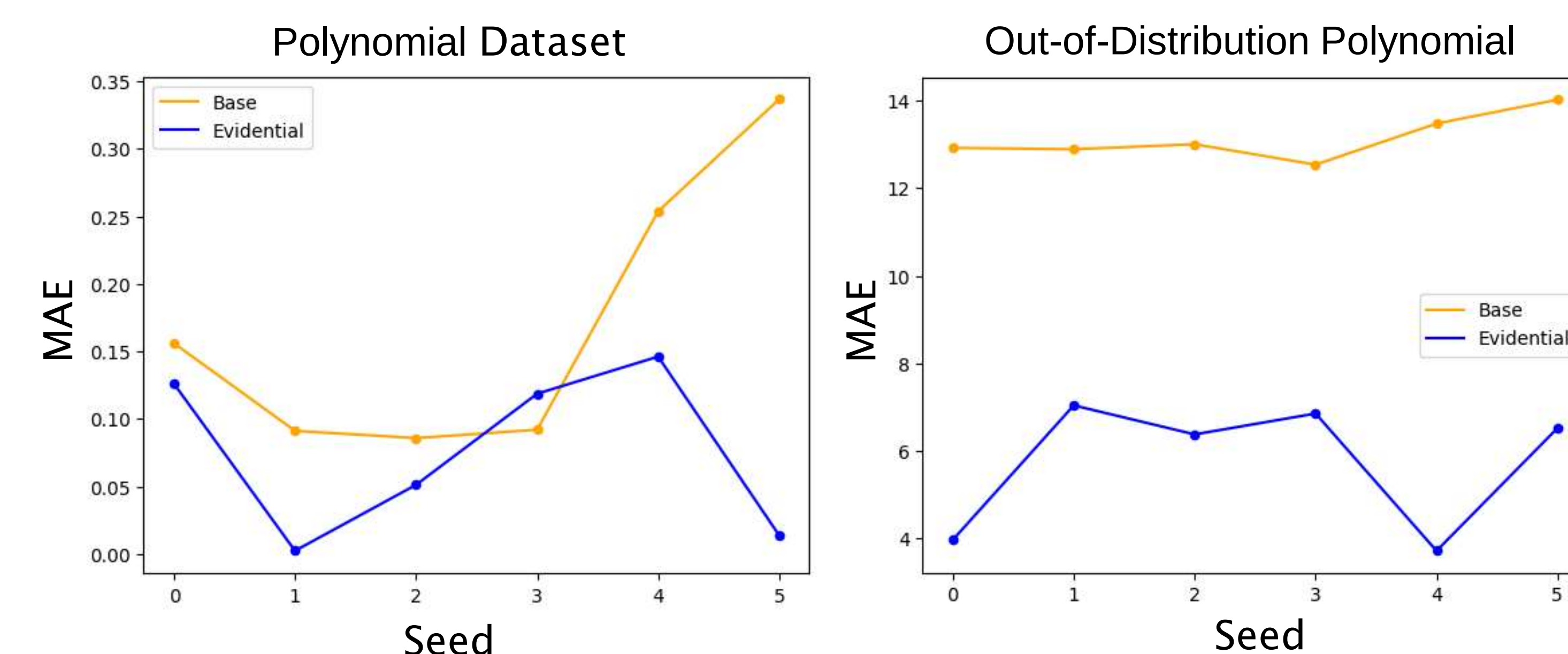
Molecule	SD	DR
	2.43	5.04
	1.77	N/A
...	...	...

Figure: Buterez et al. *chemRxiv*. (2022).

Results

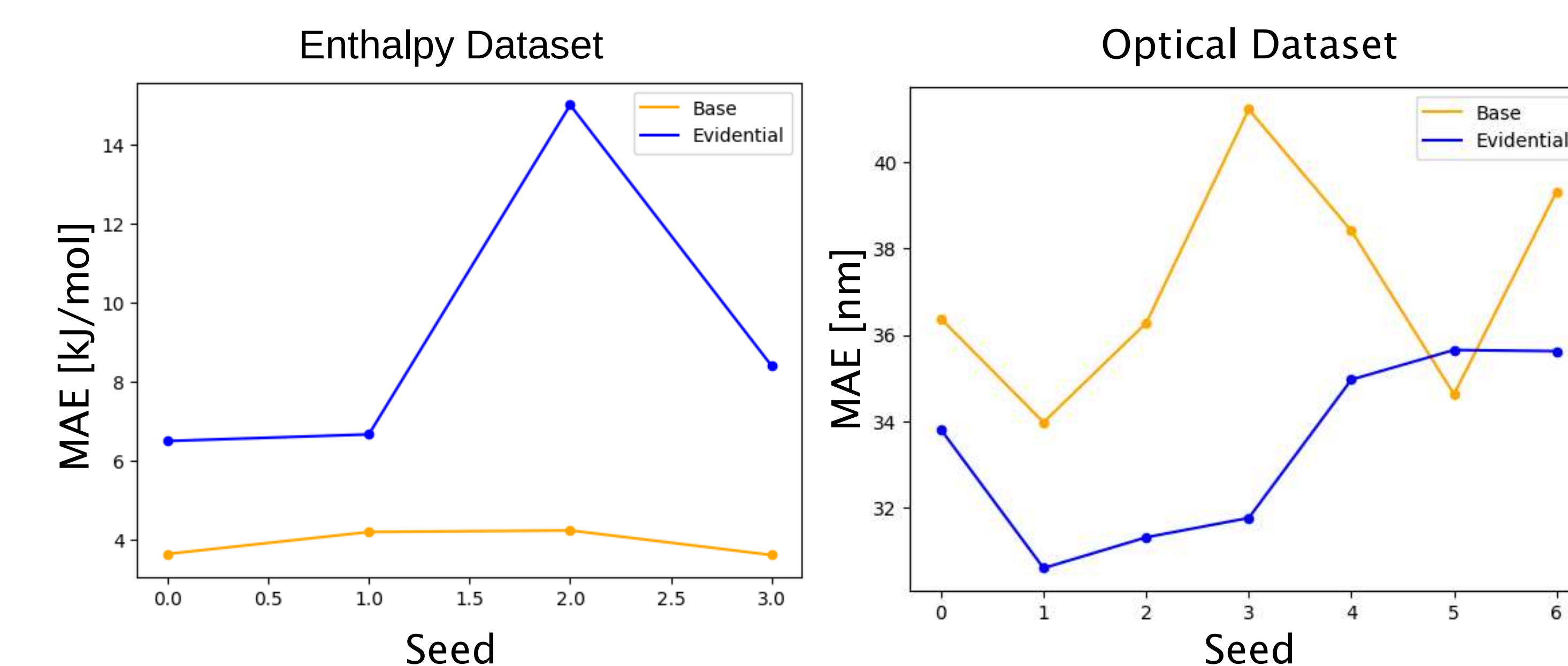
Synthetic Dataset

- Evidential neural networks generally produced lower mean average errors compared to the base multi-fidelity model
- Evidential neural networks were much more accurate in predicting out-of-distribution data.



Realistic Dataset

- Evidential neural networks generally gave lower mean average errors compared to the base multi-fidelity model.
- Evidential neural networks were slightly more stable to the variance in the train and test data.



Acknowledgements

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